Machine learning Day 1

What is Machine Learning?

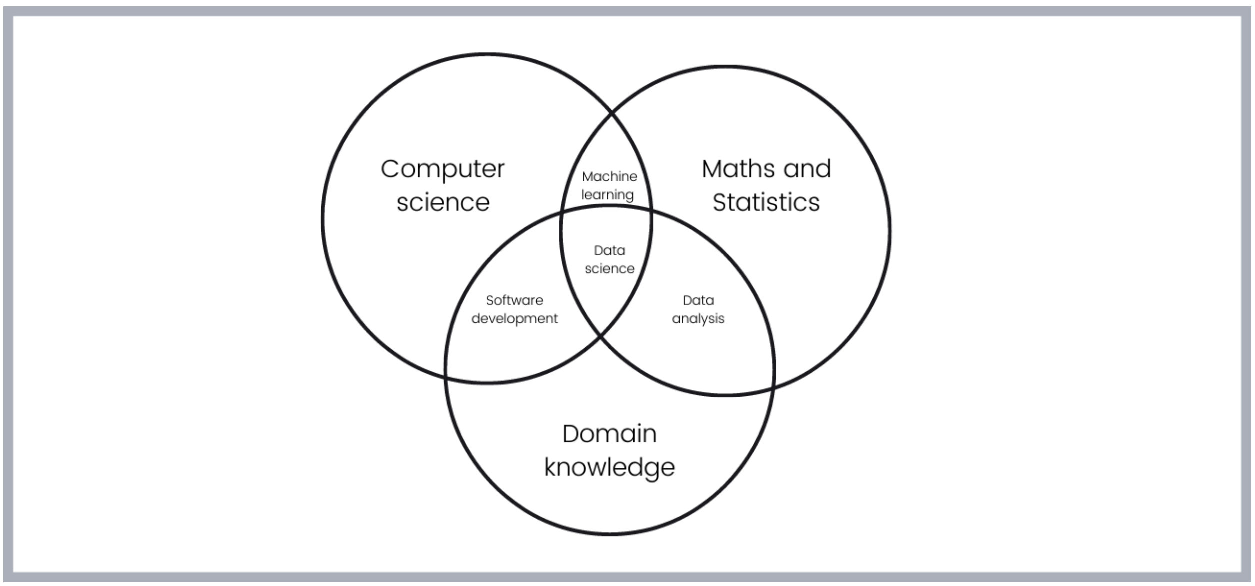
Machine learning is a new programming paradigm in which instead of explicitly programming computers to perform some tasks, we let them learn from data in order to find the underlying patterns in the data. In few words, machine learning is the science of giving the machine the ability to reason about the data.

In simple words, machine learning algorithms are trained on data rather than being programmed explicitly.

Ordinary Programming Vs Machine Learning

In ordinary programming, the job of the programmer is to clearly write every single rule that makes up the task he/she is trying to accomplish. In order to get the results, she/he must write all rules that acts up on the data.

Machine learning flips that. Instead of having to write the rules that makes up a particular application, we can feed data and results(or labels) to the machine learning model, and its job can be to determine the set of rules that map the data and labels.



Artificial Intelligence, Data Science, Machine Learning, and Deep Learning

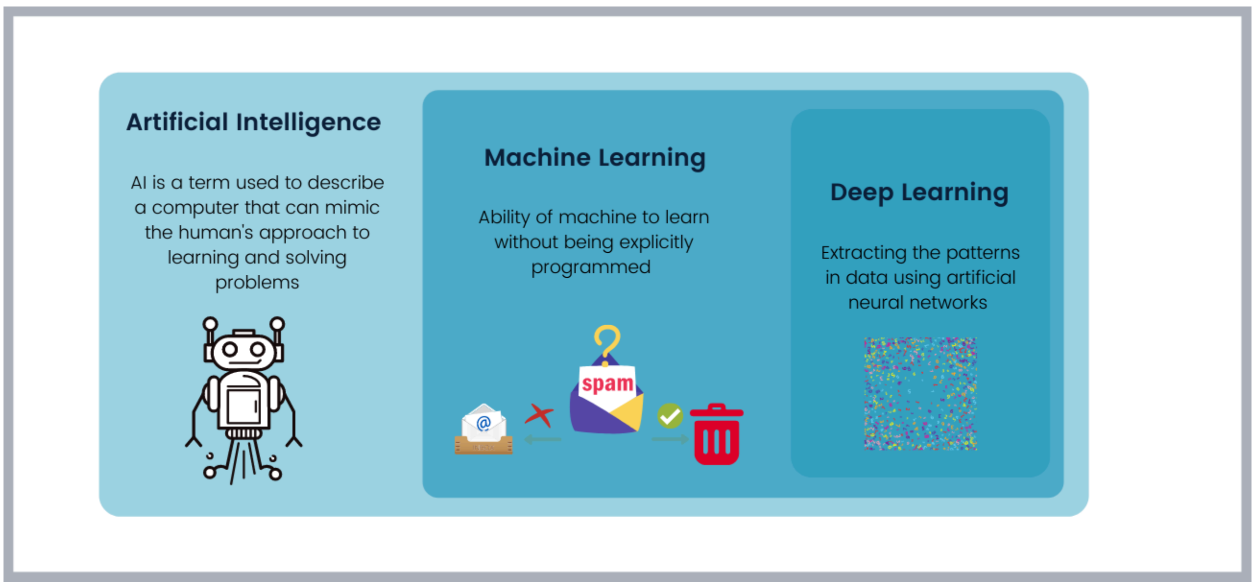
AI or Artificial intelligence, data science, machine learning, and deep learning are used interchangeably, but they are quite different.

AI is a branch of computer science concerned with building intelligent machines capable of performing tasks at the level of human. AI seeks to mimic human. AI is an interdisciplinary field that involves machine learning, programming, robotics, data science, etc...

Machine learning on the other hand is the branch of AI and as we saw, it is concerned with giving the machine the ability to learn from data. Machine learning algorithms consists of shallow or classical algorithms such as decision trees and deep learning algorithms such as convolutional neural networks

Deep learning is a branch of machine learning that deals with the study of artificial neural networks and it was inspired by the human brain. Classical machine learning algorithms needs a lot of feature engineering, but deep learning algorithms can extract features in huge amount of data such as images themselves.

Data science is also an interdisciplinary field that deals with using data to solve business problems with various techniques. A concise definition of data science was provided by Cassie Kozyrkov: *"Data Science is the science of making data useful"*.



Applications of Machine Learning

Machine learning has transformed many industries, from banking, manufacturing, streaming, autonomous vehicles, agriculture, etc...In fact, most of the tech products and services we use daily possess some sorts of machine learning algorithms running in their backgrounds.

Here are the most commonly machine learning use cases:

- **Fraud detection**: Banks and other financial organizations can use machine learning to detect frauds in real time.
- **Loan repayment prediction**: Banks can also use the historical data of their clients to predict if they will be able to repay back the loans before granting it to them.
- **Diagnosing diseases and predicting the survival rate**: Machine learning is increasingly finding its value in medicine. It can assist medical professionals in diagnozing diseases in handful of minutes. Medical professionals can also use machine learning to predict the likelihood or a course of disease or survival rate(prognosis).
- **Detecting defects in industry**: Some manufacturing companies use machine vision to inspect defects in the products which ultimately result in speeding up the production process, automating the inspection task, reducing the cost and human workload.
- **Churn prediction**: Organizations that provides some kinds of services can use machine learning to predict if a given customer is likely to opt out from the service or cancel subscription. This can help the organization to improve the customer experiences in order to retain the customers.
- **Spam detection**: Almost all email providers such as Gmail or Outlook possess the ability to detect spams from all incoming emails to protect the users from fake promotions and scams.
- **Autonomous vehicles**: Todays autonomous vehicles such as self driving cars use machine learning and deep learning systems to navigate in the roads. Using computer vision, they can be able to detect the pedestrian and traffic lights and signs and other surrounding objects.

When And When Not to Use Machine Learning

Machine learning is an incredible technology and it has shown a lot of successes in solving various real world problems. However, like any other technology, machine learning is not suitable for solving all kinds of problems. It is thus equally important to know when and when not to use machine learning.

When to use machine learning? Machine learning is preferred when approaching:

- Problems that are too complex to be solved by ordinary programming. For these kinds of problems, it's probably safe to try machine learning. There is no way one can write all rules that can be used for recognizing facemasks or detect spam emails accurately for example.
- Problems that involve visual reasoning and language understanding such as image recognition, speech recognition, machine translation, etc...As we will see later, large scale perception or visual and language problems are typically handled by deep learning systems.
- Fast changing problems where the characteristics of the problems changes with time, and there is a need to keep the system functioning well. Machine learning is suitable for these sorts of problems because machine learning algorithms can be retrained on new data.
- Problems that are clear and have simple goals such as yes/no question or predicting a single continuous number such as the quantity of product likely to be consumed in a given time.

you may not use machine learning when:

- You want the predictions made by your model to be fully explainable. This is because most machine learning models are considered to be `blackbox`.
- You do not have a reliable data for the problem you're trying to solve. A simple example here is trying to use machine learning for predicting stocks. Stock market data is unreliable and can change in a matter of seconds without any logical reasons, and so, it's pretty hard to for a model to learn some useful patterns from such unreliability.
- You can solve your problem with ordinary programming or a simple heuristic methods.
- You want a solution that will never need to be updated. The predictions made by machine learning models decay overtime, so if you are not ready to update

data and retrain models frequently, you may have to consider non machine learning techniques.

We talked about when you should use machine learning and when you should not, but also, there are other areas where machine learning is being heavily used but with extra care and human in the loop. Example of such critical areas include medicine, self driving cars, etc…In some of those areas like self driving cars(or driver assistants), machine learning is surely a big factor but also because the cost of error made by the model can be very high, human assistance becomes important.